# Neural Network Fitting Results

## Latest $\Delta P^{0.5}$ Stage-3 model, five-seed stability, and trajectory diagnostics


Linan Jiang

Aalto University


10 May 2026

# Why This Version Replaces the April Deck

## Key change

The May pipeline uses cleaner and larger upstream CDF/clean data, and the feature engineering changed from $\Delta P^{0.25}$ to the data-supported $\Delta P^{0.5}$ scale. Therefore, the 18 April tuned run is no longer the current best model, and it should not be compared directly to the retrained model as a percentage improvement.

## Strict reporting convention in this deck

- Headline results use the common external clean diagnostic: 795 561 points and 33 845 trajectories.
- The main result is the five-seed aggregate of the $\Delta P^{0.5}$ Stage-3 pipeline.
- Diagnostic plots use representative best single seed 99; these plots show error structure but do not replace the five-seed conclusion.
- The 2026-04-18 figures are archival and are not used for current model-comparison claims.

# Current Evaluation Target

| Item | Setting |
|---|---|
| Model family | Stage-3 raw-CDF refinement with $\Delta P^{0.5}$ engineered feature scaling |
| Headline protocol | external clean diagnostic, 0 ms to 5 ms |
| Headline statistic | five-seed mean $\pm$ seed-to-seed std / bootstrap CI |
| Representative plot seed | seed 99, `anchor_off`, lowest single-seed RMSE |
| Input features | $t$, tilt angle, plume count, injection duration, control backpressure |
| Removed sparse axes | direct $d_n$, $P_{\mathrm{inj}}$, $P_{\mathrm{ch}}$, and $\Delta P$ inputs |
| Target scaling | $S/A, \quad A = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}$ |

Source: `Thesis/generated/baseline_comparison_20260509/per_seed_new_mlp_eval.csv`

# Data Scale and Current Best Result

| Metric | Value |
|---|---|
| CSV files | 4,391 |
| Trajectories | 33,845 |
| Prediction points | 795,561 |
| Truth range | 0.00 mm to 95.88 mm |
| 5-seed RMSE | 6.211 mm ± 0.283 mm |
| 5-seed MAE | 4.587 mm ± 0.281 mm |
| 5-seed Bias | −0.838 mm ± 0.702 mm |
| P95 absolute error | 13.029 mm ± 0.411 mm |
| Range-normalized RMSE | 6.48% |

Source: `Thesis/generated/baseline_comparison_20260509/new_mlp_5seed_aggregate.csv`

## Representative best single model: seed 99

- RMSE $= 5.948\,\mathrm{mm}$
- MAE $= 4.383\,\mathrm{mm}$
- Bias $= -0.074\,\mathrm{mm}$
- P95 absolute error $= 12.533\,\mathrm{mm}$
- $1\sigma/2\sigma$ coverage $= 0.792\ /\ 0.985$

## Strict interpretation

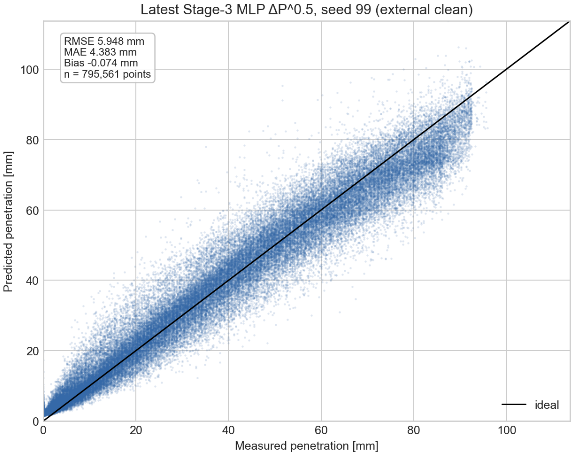
Seed 99 shows what the current pipeline can achieve in its best single run. The thesis headline should still cite the five-seed aggregate because it captures training-randomness stability.

# Metric Contract

| Metric | Meaning |
|---|---|
| RMSE | $\sqrt{\frac{1}{n}\sum_i(\hat{y}_i - y_i)^2}$; tail-sensitive primary accuracy metric. |
| MAE | $\frac{1}{n}\sum_i \lvert\hat{y}_i - y_i\rvert$; typical absolute millimetre error. |
| Bias | $\frac{1}{n}\sum_i(\hat{y}_i - y_i)$; signed systematic over-/under-prediction. |
| P95 abs. err. | 95th percentile of $\lvert\hat{y} - y\rvert$; tail-risk metric. |
| Coverage | $\Pr(\lvert y - \mu\rvert \leq k\sigma)$. Calibrated Gaussian targets are 0.683 and 0.954 for $1\sigma/2\sigma$. |

This deck therefore reports current absolute error, seed stability, tail error, and uncertainty coverage under the clean protocol; it no longer states improvement against the 4/16 baseline.

# Predicted vs. Measured Penetration

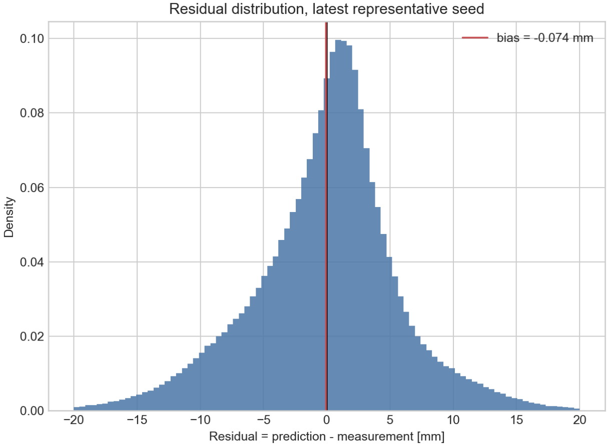



## How to read it

- Each point is one time sample in the external clean set.
- The $y = x$ line marks perfect penetration prediction.
- Seed 99 has near-zero bias, so the global under-prediction seen in older runs is strongly reduced.
- Tail scatter remains at high penetration and should be read together with P95 and trajectory review.

Source: `MLP/eval/rmse_eval_clean_20260509_215303_newdp_seed99_points/points.csv`

# Residual Distribution and Calibration

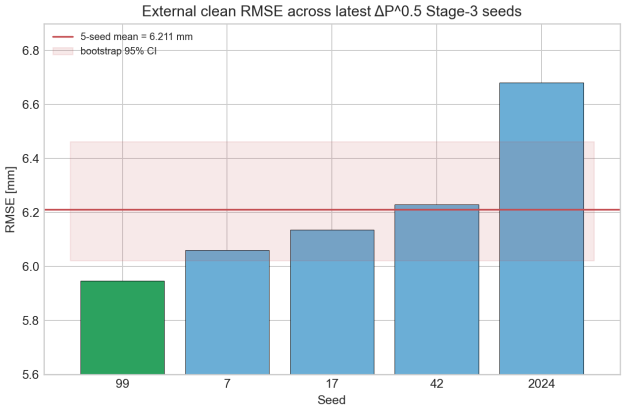



## Residual definition

$$e = \hat{y}_{\mathrm{pen}} - y_{\mathrm{pen}}.$$

- Seed 99 bias $= -0.074$ mm, close to zero global bias.
- $1\sigma$ coverage $= 0.792$, above the Gaussian nominal 0.683.
- $2\sigma$ coverage $= 0.985$, above the Gaussian nominal 0.954.
- On this clean protocol the model is conservative, which is acceptable for a screening surrogate.

# Five-Seed Stability




Source: `Thesis/generated/baseline_comparison_20260509/per_seed_new_mlp_eval.csv`

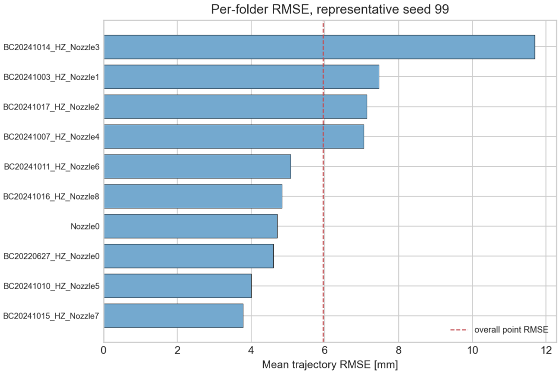
## Main readout

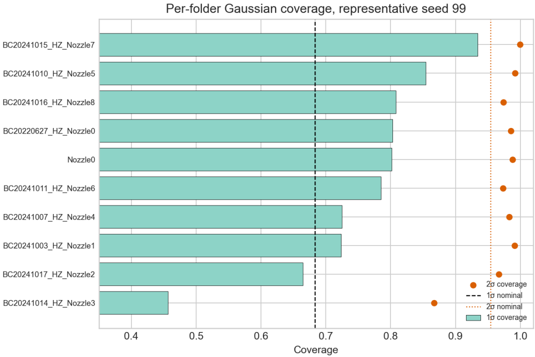
- The RMSE spread across five seeds is about $\pm 0.283$ mm.
- Seed 99 is the best single seed; seed 2024 selected a different `raw_reliable_no_kd` ablation.
- If the thesis needs strict same-ablation stability, report the four `anchor_off` seeds separately.

## Deck convention

Use the five-seed mean for the conclusion and seed 99 plots for error-structure diagnosis; they are different statistical objects.

# Per-Folder Error and Coverage

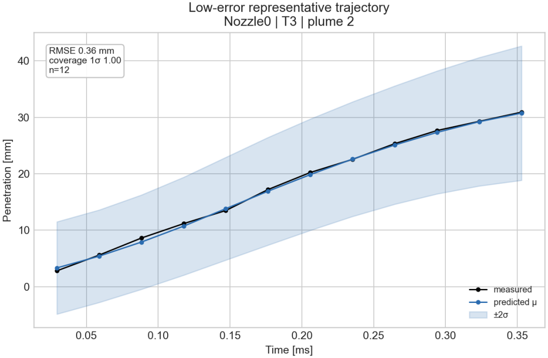


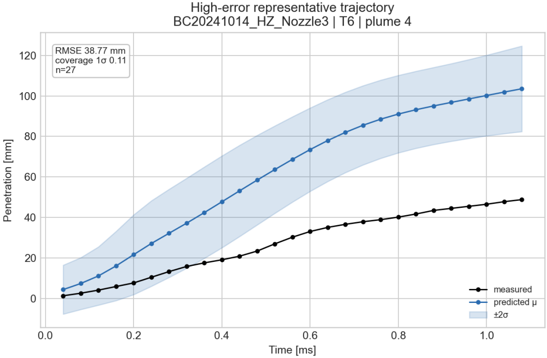

Nozzle7/5/0 are lower-error folders. BC20241014_HZ_Nozzle3 remains the hardest folder and also under-covers at $1\sigma$, making it the main trajectory-level risk point in the current model.

Source: `MLP/eval/rmse_eval_clean_20260509_215303_newdp_seed99_points/per_folder.csv`

# Representative Trajectories: Low and High Error




Nozzle0 T3 plume 2: RMSE 0.36 mm, $1\sigma$ coverage 1.00.




BC20241014_HZ_Nozzle3 T6 plume 4: RMSE 38.77 mm, $1\sigma$ coverage 0.11.

# Conclusions

1. Current reportable best result: the $\Delta P^{0.5}$ Stage-3 MLP reaches a five-seed external-clean RMSE of $6.211\,\text{mm} \pm 0.283\,\text{mm}$.
2. Latest single-seed diagnostic: seed 99 reaches RMSE $5.948\,\text{mm}$, MAE $4.383\,\text{mm}$, and bias $-0.074\,\text{mm}$, showing that global under-prediction is strongly reduced in the new pipeline.
3. Uncertainty readout: clean-set $1\sigma/2\sigma$ coverage is $0.774/0.981$ for the five-seed mean, so the model is conservative rather than under-covering.
4. Remaining risk: Nozzle3 still shows a geometry-dependent failure mode; trajectory-level review remains necessary.

**Thesis convention**

Do not compare the May retrained model directly against the April model as a percentage improvement; they use different clean data scale and a different feature-scaling protocol.

# File Index

| Content | File |
|---|---|
| Slides source | `Thesis/latex/neural_network_fit_results_slides_en.tex` |
| Compiled PDF | `Thesis/pdf/neural_network_fit_results_slides_en.pdf` |
| Latest figures | `Thesis/images/neural_network_fit_results/latest_*.png` |
| 5-seed aggregate | `Thesis/generated/baseline_comparison_20260509/new_mlp_5seed_aggregate.csv` |
| Representative seed eval | `MLP/eval/rmse_eval_clean_20260509_215303_newdp_seed99_points/` |
| Asset manifest | `Thesis/images/neural_network_fit_results/latest_assets_manifest.csv` |